

BanglaSpeechFM: Towards a Foundation Model for Bangla Speech

Anonymous ACL submission

Abstract

Self-supervised learning has greatly improved speech models, but low-resource languages like Bengali still rely heavily on large, expensive multilingual models. These large models are difficult to run and often miss the specific phonetic details of Bengali. In this paper, we introduce a Continuous Latent Autoencoder (CLAE), which is an efficient 23.8M-parameter speech foundation model built from scratch for Bengali speech. We trained it on public datasets using a new self-supervised approach that combines mel-spectrogram reconstruction, Joint-Embedding Predictive Architecture (JEPA), and Variance-Invariance-Covariance Regularization (VISReg). This allows our model to create strong representations at a 12.5 Hz latent sequence rate. Even with a small 15.29M-parameter frozen feature extractor, our model performs just as well as, and sometimes better than, much larger state-of-the-art speech models. Our results show that training a small foundation model from scratch is a practical and effective way to handle low-resource speech tasks. This provides an accessible and resource-friendly option for Bengali speech research and helps set the stage for future work in similar languages.

1 Introduction

Speech representation learning has seen remarkable progress with the advent of self-supervised learning (SSL) techniques, enabling the development of foundation models capable of understanding and processing human speech. Traditional SSL approaches in speech have often relied on reconstructing raw audio or spectrograms, which can be computationally demanding and prone to capturing low-level noise rather than high-level semantics. Recently, paradigms such as the Joint-Embedding Predictive Architecture (JEPA) have emerged as highly effective alternatives. By masking regions in high-level feature spaces and predicting latent representations rather than raw pixels or waveforms,

JEPA encourages models to learn rich, contextual semantics. While state-of-the-art (SOTA) speech models have achieved unprecedented success in high-resource languages by leveraging such techniques, low-resource languages like Bengali often suffer from a lack of dedicated, efficient models built from scratch.

Most existing works in Bengali speech AI rely on fine-tuning massive, pre-trained multilingual models, which are computationally expensive and not optimized specifically for the linguistic nuances of Bengali. Furthermore, the immense size of these SOTA models limits their deployability in resource-constrained environments. This research addresses this gap by developing an AI/ML speech foundation model specifically for Bengali, trained entirely from scratch using self-supervised learning on publicly available datasets. By doing so, we demonstrate that training a compact, dedicated model from scratch is not only viable but can also yield highly effective representations for Bengali speech.

Despite the growing interest in speech technologies, there is a significant scarcity of foundational speech models built from scratch for the Bengali language. Current approaches predominantly involve fine-tuning extremely large SOTA models trained on other languages or multilingual corpora. This reliance on massive pre-trained models presents two major challenges: first, the computational cost and memory footprint of these models make them impractical for widespread deployment or research with limited resources; second, they often lack specialized adaptation to the unique phonetic and phonological characteristics of Bengali. There are currently no researchers who have successfully developed a small-scale, highly efficient speech foundation model for Bengali from scratch using self-supervised learning on public datasets. Therefore, there is a critical need to investigate whether a significantly smaller, dedicated model

can be trained from scratch to achieve competitive or superior performance in specific metrics compared to existing heavy-weight SOTA models.

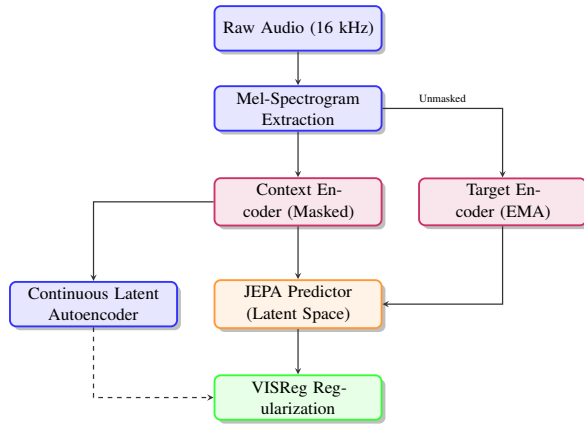


Figure 1: Proposed self-supervised CLAE framework for Bengali speech representation, integrating a Joint-Embedding Predictive Architecture (JEPA) and VISReg regularization.

To address the aforementioned challenges, our work introduces a novel learning objective that optimally combines mel-spectrogram reconstruction, JEPA-style latent space prediction, and Variance-Invariance-Covariance Regularization (VISReg), as visually outlined in Figure 1. VISReg ensures that the learned representations maintain high variance (preventing informational collapse), remain invariant to certain augmentations or masked noise, and decorrelate covariance among features, resulting in robust and disentangled speech embeddings.

The primary objective of our work is to design, train, and evaluate a Continuous Latent Autoencoder (CLAE) for Bengali speech representation learning. By developing a compact speech foundation model from scratch using self-supervised learning techniques, we aim to provide a highly efficient alternative to fine-tuning large SOTA models. The specific objectives of this research are summarized as follows:

- To develop a lightweight speech foundation model (a 23.8M-parameter continuous-latent autoencoder) specifically tailored for 16 kHz Bengali speech using publicly available datasets.
- To demonstrate that training a model from scratch using self-supervised learning is a viable and highly effective pathway for Bengali speech research, eliminating the strict dependency on fine-tuning massive multilingual

models.

- To introduce a robust self-supervised objective combining JEPA and VISReg that enhances contextual understanding and feature disentanglement without relying on raw waveform reconstruction.
- To evaluate the performance of the proposed compact model and show that despite its significantly small size compared to SOTA speech models, it achieves superior results in specific evaluation metrics.
- To provide a robust, frame-level speech representation that can be utilized for various downstream Bengali speech analysis tasks, establishing a foundation for future research in low-resource language speech AI.

2 Engines

To produce a PDF file, pdfL^AT_EX is strongly recommended (over original L^AT_EX plus dvips+ps2pdf or dvipdf). The style file acl.sty can also be used with luaL^AT_EX and XeL^AT_EX, which are especially suitable for text in non-Latin scripts. The file acl_lualatex.tex in this repository provides an example of how to use acl.sty with either luaL^AT_EX or XeL^AT_EX.

3 Preamble

The first line of the file must be

```
\documentclass[11pt]{article}
```

To load the style file in the review version:

```
\usepackage[review]{acl}
```

For the final version, omit the review option:

```
\usepackage{acl}
```

To use Times Roman, put the following in the preamble:

```
\usepackage{times}
```

(Alternatives like txfonts or newtx are also acceptable.)

Please see the L^AT_EX source of this document for comments on other packages that may be useful.

Set the title and author using \title and \author. Within the author list, format multiple authors using \and and \And and \AND; please see the L^AT_EX source for examples.

Command	Output	Command	Output
<code>\a</code>	ä	<code>\c c</code>	ç
<code>\^e</code>	ê	<code>\u g</code>	ğ
<code>\`i</code>	ì	<code>\l</code>	ł
<code>\.I</code>	İ	<code>\~n</code>	ñ
<code>\o</code>	ø	<code>\H o</code>	õ
<code>\'u</code>	ú	<code>\v r</code>	ř
<code>\aa</code>	å	<code>\ss</code>	ß

Table 1: Example commands for accented characters, to be used in, e.g., BibTeX entries.

By default, the box containing the title and author names is set to the minimum of 5 cm. If you need more space, include the following in the preamble:

```
\setlength\titlebox{<dim>}
```

where <dim> is replaced with a length. Do not set this length smaller than 5 cm.

4 Document Body

4.1 Footnotes

Footnotes are inserted with the `\footnote` command.¹

4.2 Tables and figures

See Table 1 for an example of a table and its caption.

Do not override the default caption sizes.

As much as possible, fonts in figures should conform to the document fonts. See Figure 2 for an example of a figure and its caption.

Using the `graphicx` package `graphics` files can be included within figure environment at an appropriate point within the text. The `graphicx` package supports various optional arguments to control the appearance of the figure. You must include it explicitly in the L^AT_EX preamble (after the `\documentclass` declaration and before `\begin{document}`) using `\usepackage{graphicx}`.

4.3 Hyperlinks

Users of older versions of L^AT_EX may encounter the following error during compilation:

```
\pdfendlink ended up in different nest-
ing level than \pdfstartlink.
```

This happens when pdfL^AT_EX is used and a citation splits across a page boundary. The best way to fix this is to upgrade L^AT_EX to 2018-12-01 or later.

¹This is a footnote.



Figure 2: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

4.4 Citations

Table 2 shows the syntax supported by the style files. We encourage you to use the `natbib` styles. You can use the command `\citet` (cite in text) to get “author (year)” citations, like this citation to a paper by [Gusfield \(1997\)](#). You can use the command `\citep` (cite in parentheses) to get “(author, year)” citations ([Gusfield, 1997](#)). You can use the command `\citealp` (alternative cite without parentheses) to get “author, year” citations, which is useful for using citations within parentheses (e.g. [Gusfield, 1997](#)).

A possessive citation can be made with the command `\citeposs`. This is not a standard `natbib` command, so it is generally not compatible with other style files.

4.5 References

The L^AT_EX and BibTeX style files provided roughly follow the American Psychological Association format. If your own bib file is named `custom.bib`, then placing the following before any appendices in your L^AT_EX file will generate the references section for you:

```
\bibliography{custom}
```

You can obtain the complete ACL Anthology as a BibTeX file from <https://aclweb.org/anthology/anthology.bib.gz>. To include both the Anthology and your own .bib file, use the following instead of the above.

```
\bibliography{anthology,custom}
```

Please see Section 5 for information on preparing BibTeX files.

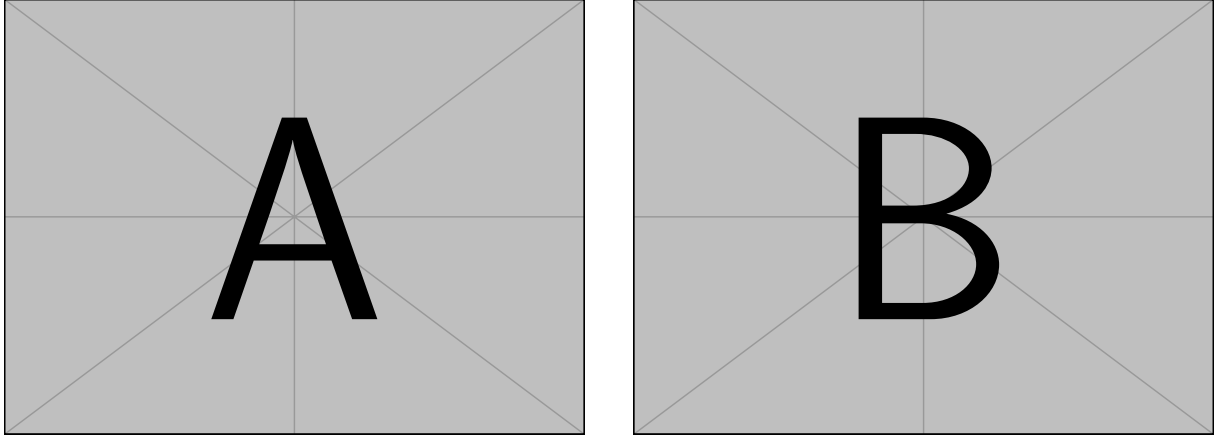


Figure 3: A minimal working example to demonstrate how to place two images side-by-side.

Output	natbib command	ACL only command
(Gusfield, 1997)	<code>\citep</code>	
Gusfield, 1997	<code>\citealp</code>	
Gusfield (1997)	<code>\citet</code>	
(1997)	<code>\citeyearpar</code>	
Gusfield’s (1997)		<code>\citeposs</code>

Table 2: Citation commands supported by the style file. The style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous ACL style files for compatibility.

4.6 Equations

An example equation is shown below:

$$A = \pi r^2 \quad (1)$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command.

This an example cross-reference to Equation 1.

4.7 Appendices

Use `\appendix` before any appendix section to switch the section numbering over to letters. See Appendix A for an example.

5 BibT_EX Files

Unicode cannot be used in BibT_EX entries, and some ways of typing special characters can disrupt BibT_EX’s alphabetization. The recommended way of typing special characters is shown in Table 1.

Please ensure that BibT_EX records contain DOIs or URLs when possible, and for all the ACL materials that you reference. Use the `doi` field for DOIs and the `url` field for URLs. If a BibT_EX entry has a URL or DOI field, the paper title in the references section will appear as a hyperlink to the paper, using the `hyperref` L^AT_EX package.

Limitations

This document does not cover the content requirements for ACL or any other specific venue. Check the author instructions for information on maximum page lengths, the required “Limitations” section, and so on.

Acknowledgments

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Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the *International Joint Conference on Artificial Intelligence* and the *Conference on Computer Vision and Pattern Recognition*.

References

Rie Kubota Ando and Tong Zhang. 2005. A framework for learning predictive structures from multiple tasks and unlabeled data. *Journal of Machine Learning Research*, 6:1817–1853.

Galen Andrew and Jianfeng Gao. 2007. Scalable training of L1-regularized log-linear models. In *Proceedings of the 24th International Conference on Machine Learning*, pages 33–40.

Dan Gusfield. 1997. *Algorithms on Strings, Trees and Sequences*. Cambridge University Press, Cambridge, UK.

Mohammad Sadegh Rasooli and Joel R. Tetreault. 2015. [Yara parser: A fast and accurate dependency parser](#). *Computing Research Repository*, arXiv:1503.06733. Version 2.

A Example Appendix

This is an appendix.